MILITARY SPECIFICATION

PRIMER, PERCUSSION, M47 AND M47B2 METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M47 and M47B2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-P-10025 — Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

MIL-STD-1167 — Ammunition Data Cards.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8838080 — Head.

8838081 — Head.

FSC 1390

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8838089 — Body.

8838012 — Cup, Battery.

8838015 — Plug, Firing.

PUBLICATIONS

ORDNANCE CORPS

IEL–7289922 — Index of Inspection Equipment Lists (47).

IEL–7299866 — Index of Inspection Equipment Lists (M47B2).

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts. The metal parts shall comply with all requirements specified on Drawings (dwgs) 8838080, 8838081, 8838089, 8838012 and 8838015 as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test (applicable to steel tubing only). The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws as, evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.4).

3.4 Workmanship. The parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time of the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

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(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agents authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractors written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 assemblies and 25 each of every individual component or every subassembly which has been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All primer bodies and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used or supply as to significantly affect product uniformity as determining by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* The components will be subjected by the Government to any or all of the examinations specified in this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the con-

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tractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, interchangeability or storage as determined by the Government, shall not necessitate changing the lot interfix number. The examinations and tests listed herein shall be performed on inspection lots submitted in accordance with MIL–STD–105, MIL–STD–1235, or MIL–STD–414 as applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL–STD–1168.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with MIL–STD–1235 may be used for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, battery (see dwg. 8838012).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Outside diameter, min.	Gage	01001
102.	Thickness through bottom, maximum (max.)	Gage	01002
103.	Length	Gage	01003
Minor:	AQL 2.50 percent		
201.	Diameter of hole	Gage	01004
202.	Thickness through bottom, min.	Gage	01005
203.	Inside diameter	Gage	01006
204.	Outside diameter, max.	Gage	01007
205.	Flat missing from surface at open end	Visual	01008
206.	Evidence of poor workmanship (see 3.4)	Visual	01009

4.3.2.2 Plug, firing (see dwg. 8838015).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Large diameter, min.	Gage	02001

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Categories	*Defects*	*Method of Inspection*	*Code No.*
102.	Length and profile of point	Gage	02002
103.	Length of pilot, min.	Gage	02003
104.	Total length	Gage	02004
Minor:	AQL 2.50 percent		
201.	Large diameter, max.	Gage	02005
202.	Length of pilot, max.	Gage	02006
203.	Diameter of pilot	Gage	02007
204.	Finish improper	Visual	02008
205.	Evidence of poor workmanship (see 3.4)	Visual	02009

4.3.2.3 Head (see dwg. 8838080 or 8838081).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 1.50 percent		
101.	Pitch diameter of thread	Gage	03001
102.	Major diameter of thread, min.	Gage	03002
103.	Diameter of body	Gage	03003
104.	Diameter of battery cup cavity	Gage	03004
105.	Diameter of percussion element cavity	Gage	03005
106.	Concentricity of flash hole with primer element cavity	Gage	03006
107.	Concentricity of flange with body diameter	Gage	03007
108.	Concentricity of percussion element cavity with body diameter	Gage	03008
109.	Thickness of flange, max.	Gage	03009
110.	Depth of battery cup cavity	Gage	03010
111.	Diameter of flash hole, min.	Gage	03011
112.	Depth of primer element cavity	Gage	03012
113.	Flash hole missing or obstructed	Visual	03013
114.	Crack or split	Visual	03014
Minor:	AQL 4.0 percent		
201.	Diameter of flange	Gage	03015
202.	Thickness of flange, min.	Gage	03016
203.	Total length	Gage	03017
204.	Length from shoulder to body end of head	Gage	03018
205.	Length from shoulder to start of thread	Gage	03019

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Categories	*Defects*	*Method of Inspection*	*Code No.*
206.	Depth of flash cavity	Gage	03020
207.	Diameter of profile of flash cavity	Gage	03021
208.	Diameter of thread pilot	Gage	03022
209.	Diameter of flash hole, max.	Gage	03023
210.	Major diameter of thread, max.	Gage	03024
211.	Diameter of thread undercut, min.	Gage	03025
212.	Length of thread undercut, max.	Gage	03026
213.	Any radii missing	Visual	03027
214.	Finish improper	Visual	03028
215.	Chamfer missing from mouth of primer element cavity	Visual	03029
216.	Evidence of poor workmanship (see 3.4)	Visual	03030

4.3.2.4 Body (see dwg. 8838089).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Flash hole missing	Gage	04001
2.	Crack or split	Visual	04002
Major:	AQL 1.00 percent		
101.	Pitch diameter of thread	Gage	04003
102.	Length of thread, min.	Gage	04004
103.	Minor diameter of thread, max.	Gage	04005
104.	Wall thickness, min.	Gage	04006
105.	Thread stripped or otherwise damaged	Visual	04007
Minor:	AQL 4.00 percent		
201.	Length of thread, max.	Gage	04008
202.	Concentricity of inside diameter with pitch diameter of thread	Gage	04009
203.	Plug gage fails to pass completely through body	Gage	04010
204.	Location of first row of holes	Gage	04011
205.	Inside diameter, min.	Gage	04012
206.	Depth of counterbore	Gage	04013
207.	Diameter of counterbore	Gage	04014
208.	Total length, max.	Gage	04015
209.	Body fails to pass through required ring gage	Gage	04016
210.	Finish improper	Visual	04017
211.	Evidence of poor workmanship (see 3.4)	Visual	04018

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4.3.3 *Testing.*

4.3.3.1 *Hydrostatic test (see 3.3) Critical defect—Code No. 04019.* This test shall be conducted 100 percent. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom. The test shall be performed as specified in 4.4.1.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL 7289922 or 7299866 as applicable identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of it's accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Hydrostatic test.* Prior to machining, the primer tubing shall be tested for not less than 5 seconds at the minimum hydrostatic pressure as calculated by the following formula.

P = (2 St times .9) divided by D

where:

P = test pressure in pounds per square inch (psi.)

S = minimum specified yield strength, psi.

t = specified minimum wall thickness in inches.

D = specified maximum outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 *Level C.* Packing and marking shall be in accordance with MIL–P–10025.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) *Data cards.* Data cards shall be prepared for each lot in accordance with the information specified in MIL–STD–1167.

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6.2 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3. Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in

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sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Unless otherwise specified on the IEL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Hydrostatic test. If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.5 Submission of initial production samples. Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NI.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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